

DEREK CHIBUZOR

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EDUCATION

University of Southern California	Los Angeles, CA
Master of Science in Mechanical Engineering	Aug. 2025-Dec. 2026
Bachelor of Science in Aerospace Engineering	Aug. 2021-May 2025

EXPERIENCE

Neros Technologies	Torrance, CA
Controls Engineer, Autonomy	Jan. 2026-Present

- * Develop performant **guidance, navigation, and control** algorithms for deployment on compute-constrained UAV platforms.
- * Contribute to internal **6-DoF flight simulator** with custom dynamics, control, state estimation, and perception modules.
- * Conduct hardware testing to tune controllers, characterize sensors, perform system identification, and evaluate software performance.

Dynamic Robotics and Control Laboratory	Los Angeles, CA
Research Assistant, Sim2Real	Jan. 2025-Present

- * Engineered **system identification** framework for **Sim2Real** transfer via gradient- and sampling-based **hybrid optimization**.
- * Trained **evolutionary algorithms** and **machine learning** models to improve **actuator dynamics** modelling by 78% in 24-DoF robot.
- * Constructed **diffusion model** framework for generating full-body **loco-manipulation** trajectories from Unitree G1 motion capture data.

Lawrence Livermore National Laboratory	Livermore, CA
Computational Engineer Intern	June 2025-Aug. 2025

- * Wrote IK task-space **PD controller** for 6-DoF UR3e, saving 5+ hours of manual path planning while achieving 0.1 mm tracking error.
- * Trained **reinforcement learning** policies for pick and place tasks, reducing episode length by 20% with transformer feature extractor.
- * Utilized implicit and explicit **structural dynamics** codes to perform **modal analysis** of jointed beam subject to broadband excitation.

Northrop Grumman	Roy, UT
Mechanical Engineer Intern	June 2023-Aug. 2024

- * Constructed 2-DoF reduced-order Simulink model of subterranean **shock-isolated system**, saving 240 hours of FEA computation.
- * Designed **CAD** assemblies, performed **FEA**, and wrote RFIs for elastomeric shock isolators and maintenance access doors.

Amazon & Information Sciences Institute	Los Angeles, CA
GNC Engineer Intern	June 2022-Aug. 2022

- * Developed **vision-based navigation** software, enabling pose estimation for 3-DoF **rendezvous and proximity operations** (RPO).
- * Achieved sub-160 ms latency localization and tracking of 12 infrared LED targets up to 2.50 m from Raspberry Pi NoIR camera.

SKILLS

Languages: Python, C++, MATLAB, Java, JavaScript
Software: Simulink, Siemens NX, PyTorch, MuJoCo, CasADi, JAX, OpenCV, LabVIEW, Betaflight
Hardware: Raspberry Pi CM5, Arduino, HIL, system identification, sensor integration